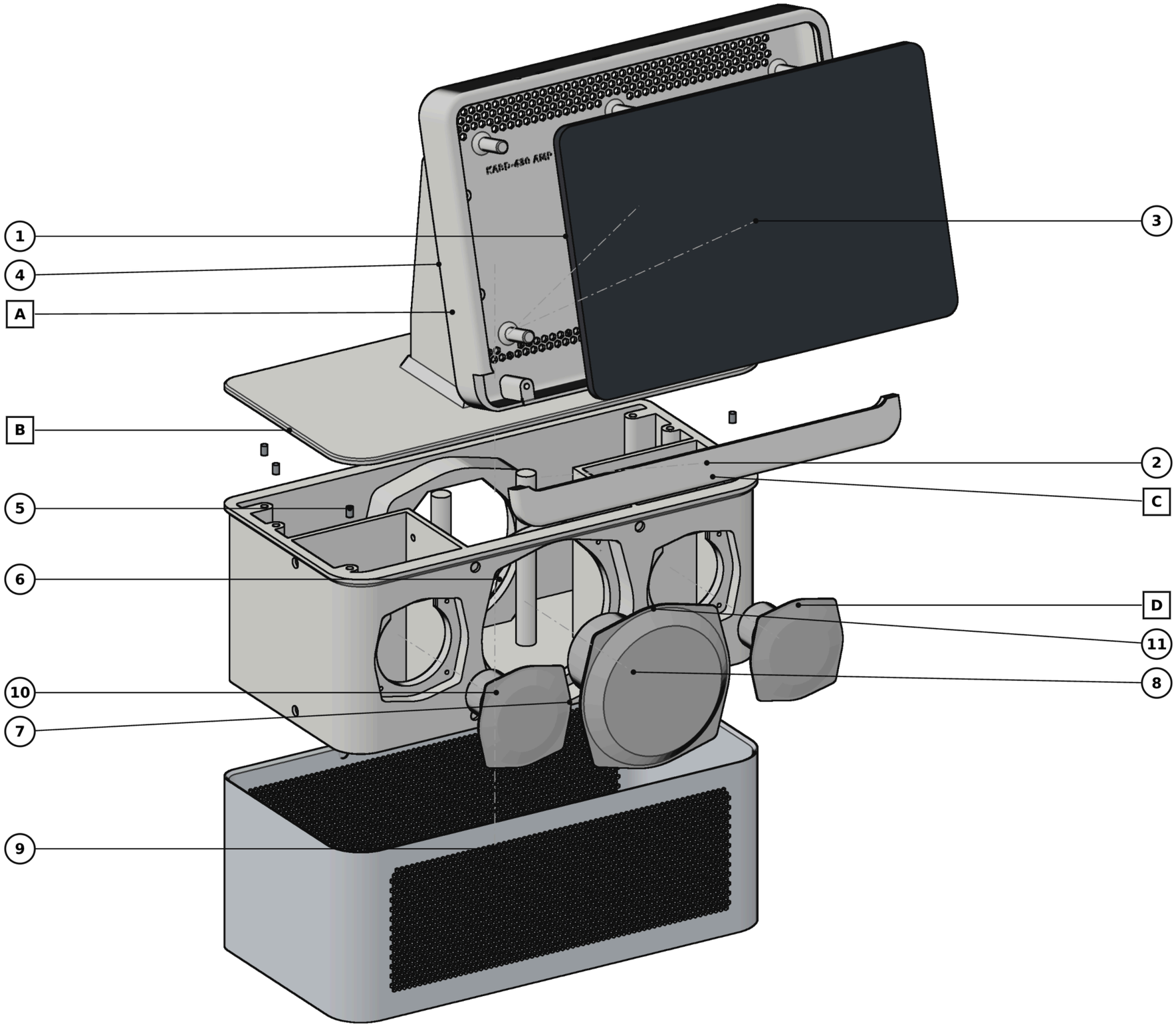


GENERAL ASSEMBLY

Exploded view, front

ITEM	DESCRIPTION	QTY	BOM
1	Bezel shell	1	P4
2	Chin bar	1	P5
3	Tablet, Galaxy Tab A9+	1	E1
4	Lid and spine	1	P2
5	Dowel pin, 4 x 6 mm	6	H5
6	Base tub	1	P1
7	Full-range driver, ND65-4	2	S1
8	Woofer, ND105-4	1	S2
9	Grille shell (or grille ring + linen)	1	P3a
10	Driver screw, M3 x 6 thread-forming	16	S4
11	Gasket tape, 1 mm (under each flange)	1	S5

NOTES

- **A** Bezel bolts to the spine from the rear (sheet 2).
- **B** Epoxy the lid to the tub: outer rim and both full-range chamber walls. Airtight.
- **C** Tablet slides up into the shell from below, then the chin bar closes the slot.
- **D** ND65: tabs up, through the opening above the cutout. Solder after mounting, then seal the opening from inside.

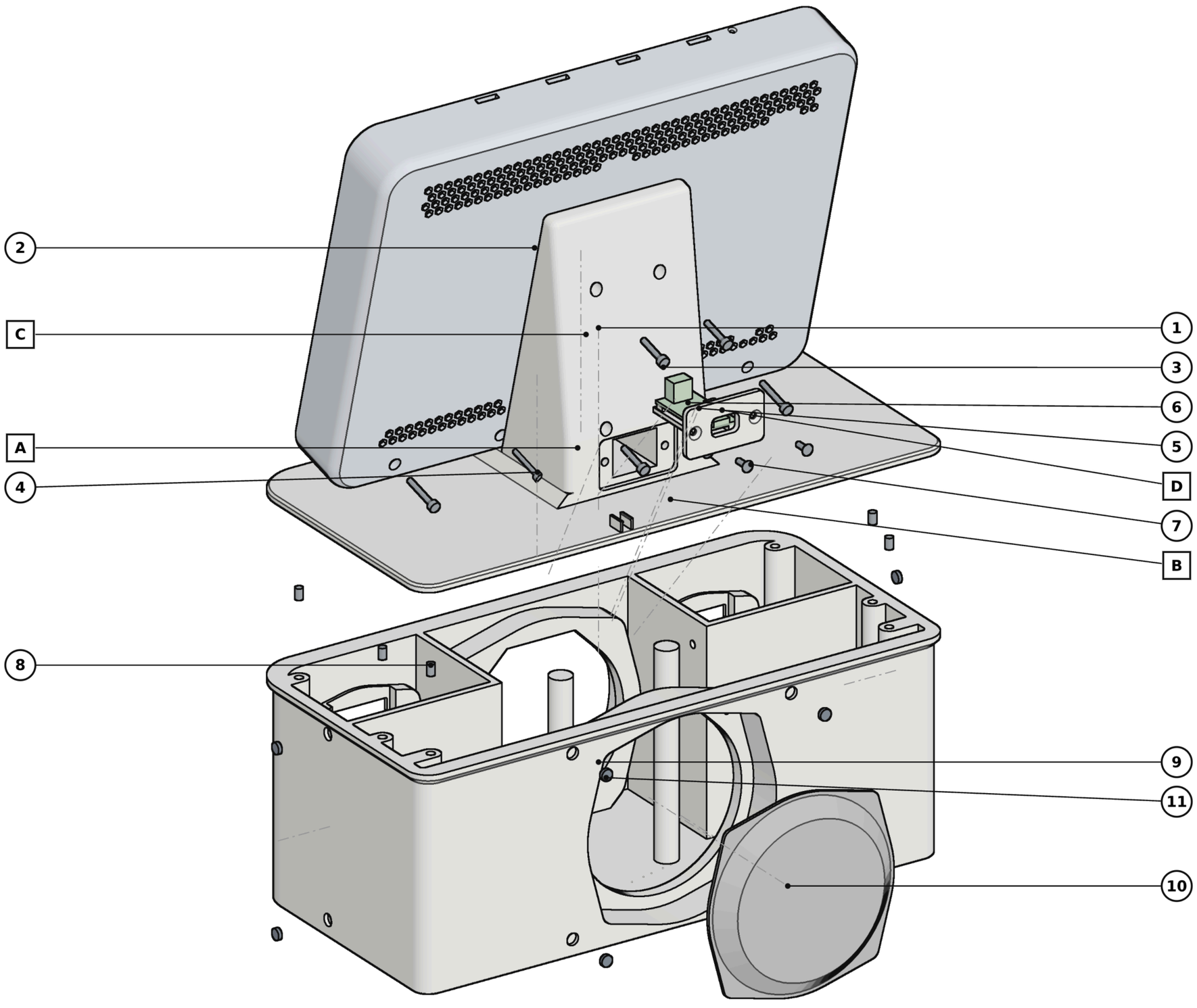
SMARTHUB · COUNTER DISPLAY

DRAWING
GENERAL ASSEMBLY

DWG NO. SH-ASM-01	SHEET 1 OF 4
REV A	DATE 2026-11-10
SCALE NTS	SOURCE enclosure.py d29037f

BASE AND SPINE

Exploded view, rear



ITEM	DESCRIPTION	QTY	BOM
1	Lid and spine	1	P2
2	Bezel assembly (sheet 3)	1	
3	Bezel bolt, M3 x 16 socket head	4	H3
4	Chin bar screw, M3 x 20 socket head	4	H2
5	Power plate	1	P6
6	PD trigger, HUNSOOL (DIP = 20 V)	1	E9
7	Plate screw, M3 x 8 countersunk	2	H3a
8	Dowel pin, 4 x 6 mm	6	H5
9	Base tub	1	P1
10	Passive radiator, ND105-PR	1	S3
11	Magnet, 6 x 2 mm (tub pockets)	12	H6

NOTES

A. Ø8 speaker wire hole in the lid, under the spine. After the lid is glued on, seal it with silicone from above through the empty power plate pocket.

B. Cable clip: press the charger cable in here.

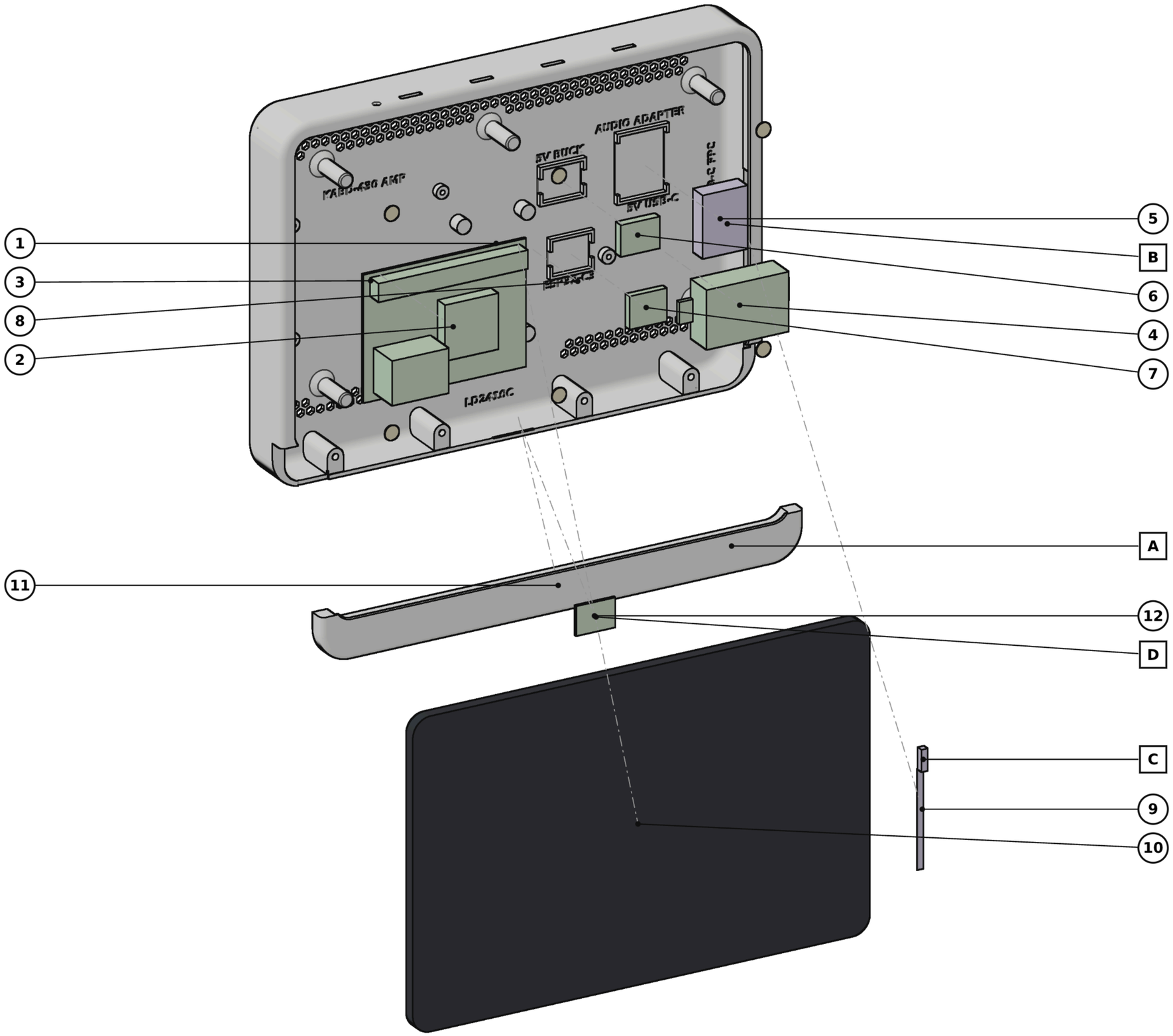
C. Speaker and 20 V leads run up the spine channel into the tray.

D. Set the DIP switch to 20 V only before fitting the plate.

SMARTHUB · COUNTER DISPLAY

DRAWING
BASE AND SPINE

DWG NO.	SHEET
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NTS	enclosure.py d29037f



BEZEL ASSEMBLY

Exploded view, front (bezel frame, upright)

ITEM	DESCRIPTION	QTY	BOM
1	Bezel shell	1	P4
2	Amplifier, Dayton KABD-430	1	E2
3	Amp screw, M3 x 6 thread-forming	4	H4
4	5 V USB-C converter (charges the tablet)	1	E6
5	UGREEN 2-in-1 USB-C audio + PD adapter	1	A2
6	5 V buck, MP1584EN	1	E5
7	Seeed XIAO ESP32-C3	1	E3
8	Felt pad, 8 mm, 0.5 mm thick	6	H9
9	Flat FPC USB-C 90° adapter	1	A1
10	Tablet, Galaxy Tab A9+	1	E1
11	Chin bar	1	P5
12	Radar, HLK-LD2410C	1	E4

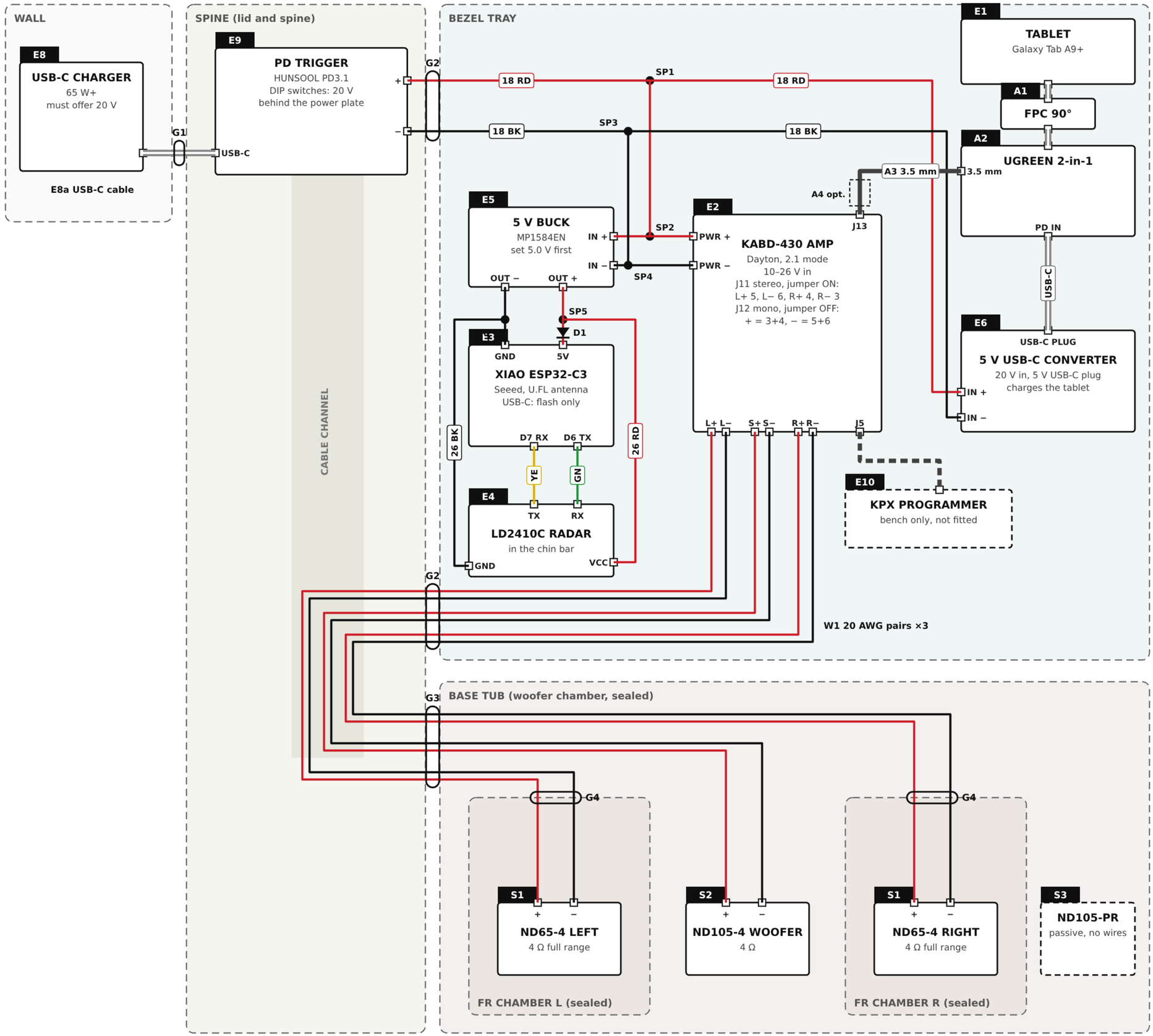
NOTES

- **A** Heat-set inserts: 4 in the chin bar's back face, 4 in the spine-bolt bosses (from the back).
- **B** Small boards sit in their labeled cradles on VHB tape.
- **C** Plug the FPC adapter into the tablet first. It rides in the channel on the right as the tablet slides up into the shell.
- **D** Radar looks out through 1.6 mm of the chin. No metal-filled filament.

SMARTHUB · COUNTER DISPLAY

DRAWING
BEZEL ASSEMBLY

DWG NO.	SHEET
SH-ASM-03	3 OF 4
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SCALE	SOURCE
NTS	enclosure.py d29037f



WIRING DIAGRAM

Power, audio and sensor circuits

ID	COMPONENT	WHERE
E8	USB-C PD charger, 65 W+	wall
E9	PD trigger, HUNSOOL (DIP 20 V)	spine
E2	KABD-430 amplifier	tray
E6	5 V USB-C converter	tray
E5	5 V buck, MP1584EN	tray
E3	Seeed XIAO ESP32-C3	tray
E4	LD2410C radar	chin
A1 / A2	FPC 90° adapter / UGREEN 2-in-1	tray
A3 / A4	3.5 mm cable / ground loop isolator (opt.)	tray
S1–S3	ND65-4 ×2, ND105-4, ND105-PR	tub
E10	KPX programmer (to J5, bench only)	—

WIRE	CODE	USE
	18 RD	+20 V (W2)
	18 BK	Ground (W2)
	20 pair	Speaker, + / − (W1)
	26 RD / BK	5 V / ground (W3)
	26 YE / GN	Radar TX → D7 (IO20), RX ← D6 (IO21)
	USB-C	Cables and adapters
	3.5 mm	Line-level audio

PASS-THROUGHS

- **G1** Power plate opening on the spine's rear.
- **G2** Ø13 cable hole in the tray's back wall, at the top of the spine channel.
- **G3** Ø8 lid hole into the woofer chamber. **Fill with hot glue.**
- **G4** Full-range chamber wall holes ×2. **Fill with hot glue.**

NOTES

1. Check for 20 V at E9's terminal before connecting anything.
2. If the buck is adjustable, set it to 5.0 V before connecting E3 and E4.
3. J11 runs stereo for the full-range pair. J12 runs mono (bridged) for the woofer. The crossover comes from a 2.1 DSP program loaded through J5. J13 is the AUX input: 5-pin JST PH, 1 L, 2 AGND, 3 R. AUX and Bluetooth are mixed by default. Change the mode jumpers only with the power off.
4. Speaker + goes to each driver's red-marked terminal.
5. Fit A4 only if you hear hum while the tablet charges.
6. D1 is a Schottky diode (1N5819 or SS14) into the XIAO's 5V pin, so USB can't back-feed the buck. Plug in the XIAO's antenna.

SMARTHUB · COUNTER DISPLAY

DRAWING
WIRING DIAGRAM

DWG NO. SH-WIR-01	SHEET 4 OF 4
REV A	DATE 2026-11-05
SCALE NTS	SOURCE enclosure.py f156b7e